

Calendar — Practice Questions

Medium to Difficult, With Full Solutions

Aptitude Grind

Difficulty is marked as **[M]** Medium or **[H]** Hard. Solve on your own first, then check the solution. Solutions use the odd-days method from *01-fundamentals.tex*.

Q1 [M] — Direct Date Lookup

What was the day of the week on 26 January 1950?

★ Solution

Reference: 1 January 1900 was a **Monday**.

- Years 1900–1949 (50 years elapsed up to start of 1950... but we need up to 1 Jan 1950): count odd days for years 1900 to 1949 inclusive of leap years.
- Leap years 1900–1949: 1904, 1908, ..., 1948 — but 1900 is *not* leap (century rule) $\Rightarrow$ 12 leap years, 38 ordinary years.
- Odd days = $(12 \times 2) + (38 \times 1) = 24 + 38 = 62 \equiv 62 - 56 = 6 \pmod{7}$
- Monday +6 odd days $\rightarrow$ **Sunday** (this gives the day for 1 Jan 1950)
- Add 25 more days to reach 26 Jan: $25 \bmod 7 = 4$ odd days
- Sunday +4 $\rightarrow$ **Thursday**

Answer: Thursday

Q2 [M] — N Days After

Today is Wednesday. What day will it be after 90 days?

★ Solution

$90 \bmod 7 = 90 - 84 = 6$ odd days. Wednesday +6 days $\rightarrow$ **Tuesday**. **Answer: Tuesday**

Q3 [M] — Same Date, Different Year

If 15 March 2015 was a Sunday, what day was 15 March 2016?

★ Solution

2015 is an ordinary year, but the gap 15 Mar 2015 → 15 Mar 2016 **includes 29 Feb 2016** (since 2016 is a leap year and the leap day falls before 15 March). So this interval has **2 odd days**, not 1. Sunday +2 → **Tuesday**. **Answer: Tuesday**

Why This Trips People Up

The "add 1 odd day per year" shortcut only works when the year in question is ordinary **and** the leap day (if any) doesn't fall between the two matching dates. Always check whether 29 February lies inside the specific date range you're bridging — not just whether the calendar year is leap.

Q4 [M] — Same Calendar Year

The calendar for the year 2023 will be the same as the calendar for which future year?

★ Solution

2023 is ordinary. Track running odd days year by year starting the count from 2023→2024:

Transition	Odd days added	Running total (mod 7)
2023 → 2024 (2023 ordinary)	1	1
2024 → 2025 (2024 leap)	2	3
2025 → 2026 (2025 ordinary)	1	4
2026 → 2027 (2026 ordinary)	1	5
2027 → 2028 (2027 ordinary)	1	6
2028 → 2029 (2028 leap)	2	1
2029 → 2030 (2029 ordinary)	1	2
2030 → 2031 (2030 ordinary)	1	3
2031 → 2032 (2031 ordinary)	1	4
2032 → 2033 (2032 leap)	2	6
2033 → 2034 (2033 ordinary)	1	0

Running total first hits a multiple of 7 at **2034**, and 2034 is also an ordinary year (matching 2023's type). **Answer: 2034**

Q5 [H] — Century Reasoning

Prove that the last day of a century (year ending in 00) can never be a Tuesday, Thursday, or Saturday.

★ Solution

The last day of the n -th century corresponds to a cumulative odd-day count from a fixed starting point equal to $5n \bmod 7$ for n not a multiple of 4, and 0 when n is a multiple of 4 (from the 400-year rule). Evaluate $5n \bmod 7$ for $n = 1, 2, 3$ (since $n = 4$ gives 0):

$$n = 1 : 5 \quad n = 2 : 10 \equiv 3 \quad n = 3 : 15 \equiv 1 \quad n = 4 : 0$$

So the only possible odd-day residues for a century's last day are $\{0, 1, 3, 5\}$, which — using the reference that year 1's last day (31 Dec, year 0 effectively) corresponds to Sunday being residue 0 — map to a fixed subset of just four weekdays. Direct computation (as shown in standard references) confirms centuries can only end on **Sunday, Monday, Wednesday, or Friday**, never Tuesday, Thursday, or Saturday. **Answer: Because only 4 distinct odd-day residues (0, 1, 3, 5) are possible for century boundaries, only 4 of the 7 weekdays are reachable — and Tuesday, Thursday, Saturday are never among them.**

Q6 [H] — Reverse Problem

1 January of a certain year is a Friday. 1 January of the following year is a Sunday. What can you conclude about the certain year, and what day is 1 January two years later?

★ Solution

Going from Friday to Sunday is a shift of **2 odd days** $\Rightarrow$ the certain year must be a **leap year** (only leap years contribute 2 odd days). So the certain year has 366 days, and the following year is ordinary, contributing 1 odd day. Sunday +1 $\rightarrow$ **Monday** for 1 January two years later. **Answer: The certain year is a leap year; 1 January two years later is a Monday.**

Q7 [H] — Multi-Step Combined Problem

If 1 January 2001 was a Monday, find the day of the week on 29 February 2004, and verify 2004 is indeed a leap year using the divisibility rule.

★ Solution

Leap year check: $2004/4 = 501$ exactly, and 2004 is not a century year, so **2004 is a leap year**. ✓

Odd days from 1 Jan 2001 to 1 Jan 2004 (years 2001, 2002, 2003 — all ordinary): $3 \times 1 = 3$ odd days. Monday +3 $\rightarrow$ **Thursday** (this is 1 Jan 2004).

Odd days from 1 Jan 2004 to 29 Feb 2004: January has 31 days $\rightarrow 31 \bmod 7 = 3$ odd days (this brings us to 1 Feb). Add 28 more days to reach 29 Feb: $28 \bmod 7 = 0$ odd days. Total: $3 + 0 = 3$ odd days.

Thursday +3 $\rightarrow$ **Sunday**. **Answer: 29 February 2004 was a Sunday.**

Q8 [H] — Century-Spanning Problem

If 1 January 2001 was a Monday, what day of the week is 1 January 2101?

★ Solution

We need the total odd days contributed by the years 2001 through 2100 (100 full years elapsing).

Leap years in 2001–2100: every year divisible by 4 is a leap year *unless* it's a century year not divisible by 400. In this range: 2004, 2008, ..., 2096 are leap $\Rightarrow \frac{2096 - 2004}{4} + 1 = 24$

leap years. The only century year in range, 2100, is **not** divisible by 400, so it is *not* leap (and it's already excluded from the count above since it isn't divisible by 4 in the "leap" sense we need to re-verify — 2100 is divisible by 4 but excluded by the century rule, so it contributes as an **ordinary** year instead).

- Leap years: 24
- Ordinary years: $100 - 24 = 76$
- Odd days = $(24 \times 2) + (76 \times 1) = 48 + 76 = 124$
- $124 \bmod 7 = 124 - 119 = 5$

Monday +5 odd days → **Saturday**. **Answer: 1 January 2101 is a Saturday.**

Why This Is a Common CAT/Placement Trap

The trap is treating 2100 as a normal leap year because it's divisible by 4. Since it's a century year **not** divisible by 400, it must be excluded — this single oversight shifts the final answer by one full odd day (which changes the answer from Saturday to Sunday if missed).

Answer Key Summary

Question	Answer
Q1	Thursday
Q2	Tuesday
Q3	Tuesday
Q4	2034
Q5	Only Sun/Mon/Wed/Fri possible for century-end (proof-based)
Q6	Certain year is leap; 2 years later = Monday
Q7	Sunday
Q8	Saturday